



INT303 BIGDATA ANALYSIS

COURSEWORK 2

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BSC INFORMATION MANAGEMENT AND INFORMATION SYSTEMS

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1. Introduction

This project develops a predictive modelling framework for loan approval using financial, demographic, and credit-related attributes of applicants. Formulated as a binary classification task,

$$y_i \in \{0, 1\}, \quad \hat{y}_i = f(\mathbf{x}_i; \theta),$$

the objective is to design a model that achieves high predictive accuracy, interpretability, and operational feasibility for real-world FinTech decision-support systems.

2. Methodology

2.1 Data Processing

Missing numerical values were imputed using medians, while categorical attributes were imputed using modes. Categorical variables were transformed using one-hot encoding, and all numerical inputs were standardised to ensure comparability across heterogeneous financial variables. This preprocessing pipeline stabilises optimisation behaviour across models and maintains meaningful variance within applicant characteristics.

2.2 Feature Engineering

Two financially meaningful features were engineered to enhance representational richness. The first is the debt-to-income ratio,

$$\text{DTI} = \frac{\text{loan_amount}}{\text{income_annum}},$$

which captures the repayment burden relative to earning capacity. The second is an aggregated asset indicator,

$$A_{\text{total}} = A_{\text{res}} + A_{\text{comm}} + A_{\text{lux}} + A_{\text{bank}},$$

providing a consolidated measure of financial stability. These features improve the model's ability to characterise repayment capacity and overall borrower resilience.

2.3 Models and Hyperparameter Tuning

Six supervised learning models were trained: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, SVM (RBF kernel), and KNN. Hyperparameter optimisation was conducted using GridSearchCV with 5-fold cross-validation, maximising the F1-score,

$$\text{F1} = \frac{2PR}{P+R},$$

which is appropriate given the moderate class imbalance present in the dataset. Tuning controlled model capacity and regularisation strength, yielding substantial improvements in generalisation for most models.

3. Key Findings

3.1 Model Performance

All baseline models performed strongly, but ensemble methods were dominant. Before tuning, Random Forest and Gradient Boosting already achieved accuracies above 0.97 and ROC AUC values exceeding 0.99. After hyperparameter optimisation,

these models further improved, with tuned Gradient Boosting and Decision Tree both achieving approximately 0.98 accuracy and ROC AUC values around 0.998–0.989. Logistic Regression improved its F1-score for the approved class from approximately 0.87 to 0.92, while SVM improved from roughly 0.90 to 0.92. KNN showed moderate gains but remained the least competitive overall. These enhancements illustrate the importance of controlling variance, margin behaviour, and neighbour-based decision boundaries.

3.2 Feature Importance

Feature-importance analysis from the tuned Random Forest and Gradient Boosting models demonstrates that credit score overwhelmingly dominates predictive influence:

$$\text{Imp}(\text{cibil_score}) \approx 0.81 \text{ (RF)}, \quad \text{Imp}(\text{cibil_score}) \approx 0.87 \text{ (GB)},$$

far exceeding the contributions of all other attributes. Loan term and loan amount consistently ranked within the next most influential features, reflecting heightened long-term exposure and capital risk. Income- and asset-related variables such as *income_annum*, *residential_assets_value*, and *bank_asset_value* contributed moderately. Demographic variables, including number of dependents, contributed minimally, indicating that the models rely primarily on economic fundamentals rather than demographic characteristics—an important consideration for fairness in credit assessment.

4. Recommendations & Insights

4.1 Recommended Model

The Gradient Boosting Classifier is recommended for operational deployment due to its superior Accuracy (≈ 0.98), strong class-specific F1-scores (both exceeding 0.96), and consistently high ROC AUC (≈ 0.998). Its probabilistic outputs may be mapped to lending decisions using

$$\hat{y} = \begin{cases} 1, & \hat{p} \geq \tau, \\ 0, & \hat{p} < \tau, \end{cases}$$

where the threshold τ can be adjusted in accordance with institutional risk tolerance or regulatory constraints.

4.2 Business Insights

The empirical findings reinforce established lending principles. Strong credit history (high CIBIL score) is the primary determinant of approval outcomes, while higher income and asset positions correlate with lower default risk. Larger loan amounts and longer repayment horizons increase financial exposure, elevating borrower risk. Engineered variables such as DTI and total assets enhanced predictive clarity, enabling the model to capture structural aspects of repayment capacity.

5. Ethical Considerations

Although demographic attributes exhibited minimal influence, fairness remains essential. Disparate impact may be evaluated using

$$\text{DI} = \frac{\Pr(\hat{y} = 1 \mid g = 1)}{\Pr(\hat{y} = 1 \mid g = 0)},$$

to detect unintended disparities. For transparency and regulatory alignment, SHAP-based explanations are recommended to provide interpretable, instance-level justification for automated lending decisions.